FastSpeech 2-Based Text-to-Speech Synthesis

Experimental Implementation Report: English and Telugu

1. PROJECT OVERVIEW

This project investigates a FastSpeech 2-based text-to-speech system for English and Telugu using the reference implementation. This is a partial experimental implementation, not a completed bilingual speech synthesis system.

2. ENVIRONMENT AND MODEL

Experiments were conducted in Kaggle using Python 3.13.16, PyTorch 2.11.0 with CUDA 13.0, and an NVIDIA Tesla T4 GPU. The model used an encoder hidden size of 256, decoder hidden size of 256, four encoder layers, six decoder layers, and 80-channel mel-spectrogram representations.

3. ENGLISH DATASET AND PREPROCESSING

The English experiment used the Svarah dataset and its available preprocessed features. Inspected metadata contained 3,341 training entries and 675 validation entries, with separate training and validation speaker directories.
Features included mel-spectrograms, duration, pitch, and energy. The configuration used 22,050 Hz audio, 1024-point FFT, 256-sample hop length, and 80 mel channels.

Training and Inference Results

4. ENGLISH TRAINING EXPERIMENT

An experimental run was performed for 20 training steps with

a batch size of 8. Checkpoints were saved at steps 10 and 20.

The loss values fluctuated, including a large spike at step 3.

This short experiment does not establish convergence.

Recorded training losses:

Step 01: 17.8609

Step 02: 13.4417

Step 03: 134.1252

Step 04: 43.7589

Step 05: 15.8959

Step 06: 7.2528

Step 07: 7.6589

Step 08: 6.6609

Step 09: 7.3918

Step 10: 9.9467

Step 11: 11.4781

Step 12: 9.0254

Step 13: 9.0529

Step 14: 8.9433

Step 15: 7.0096

Step 16: 6.8858

Step 17: 5.9080

Step 18: 5.4780

Step 19: 6.5231

Step 20: 6.5494

5. ENGLISH INFERENCE OBSERVATION

One English test utterance (E2a) was synthesized and saved

as a WAV file. Its duration was approximately 6.97 seconds

at 22,050 Hz. Listening indicated a continuous tone-like sound

rather than intelligible speech. The current checkpoint

therefore does not provide satisfactory speech synthesis.

6. TELUGU PREPARATION

Telugu Unicode symbol support and tokenization were tested.

The vocabulary contained 458 symbols, and a sample sentence

could be tokenized and reconstructed. However, valid Telugu

acoustic alignments, model training, checkpoint creation,

and speech inference were not completed.

Limitations and Future Work

7. VALIDATION AND VARIANCE ADAPTOR ANALYSIS

A validation-loss experiment was not completed. No validation loss curve or convergence claim is therefore provided.
The required per-test-case duration, pitch, and energy variance adaptor plots were also not generated.

8. LIMITATIONS AND FUTURE WORK

The English training run needs to be extended substantially, followed by validation and systematic inference testing.
English pronunciation handling and generated audio quality require improvement.

Telugu requires a suitable text-to-phoneme pipeline and reliable acoustic alignment before model training.
Future work should complete both language pipelines, generate all required test utterances, inspect duration, pitch and energy predictions, and compare both languages.

9. SUBMISSION STATUS

The included English checkpoints correspond to the 20-step experimental run. The repository does not contain a completed Telugu model or a fully validated bilingual TTS system.

This report records the work actually performed and its current limitations. No uncompleted results are claimed.